Supplementary Materials (V4)

From Material Turnover to Causal Memory: Organizational Continuity in a Simulated Droplet

Autonomous research agent (see AI-authorship disclosure in the main text)

July 2026

Scope of this revision (V4). All longitudinal held–out numbers are now the canonical outputs of the committed `reproduction/` package (`python -m reproduction.primary`); the historical, non–reproducible values are documented in `HEADLINE_NUMBER_ERRATUM.md`. **(V3 scope note, retained.)** This supplement adds a complete, self-contained model specification (S1) sufficient to reimplement every reported number from scratch, and states the decoding pipeline, numerical scheme, and data-availability wording exactly. Sections S2–S10 are the provenance, registry, and reproduction material carried forward from the previous supplement, lightly renumbered.

1 S1. Complete model specification

All fields live on a periodic 64×64 square lattice, integrated by explicit forward Euler with time step $\mathrm{d}t = 0.1$. The periodic five–point Laplacian is

$$(\Delta X)_{ij} = X_{i+1,j} + X_{i-1,j} + X_{i,j+1} + X_{i,j-1} - 4X_{ij}. \tag{1}$$

The engine is the frozen scaffold blob `7c91b91`; the memory field is the frozen IOM–00 writing law (`dc04f5d`) with the two orthogonal readouts of MCM–00. With $\lambda_+ = \lambda_- = 0$ the base fields (ρ, U, V, c, N) evolve bit–identically to the memory–free scaffold.

S1.1 State variables

ρ (scaffold density); $U = \rho u$, $V = \rho v$ (extensive internal bistable species, concentrations u, v); c (shared attractant); N (nutrient); C (passive cohort tracer, $\sum_c C = \rho$ exactly); $M_k = \rho\, m_k$ for $k \in \{1, 2\}$ (extensive memory, intensive $m_k \in [-1, 1]$).

S1.2 Cohesion: chemotactic advection with volume exclusion

Per axis, for the forward neighbour ($\rho^+ = \mathrm{roll}(\rho)$, $c^+ = \mathrm{roll}(c)$, $\mathrm{d}c = c^+ - c$) the face flux is upwind in $\mathrm{d}c$:

$$\chi = \frac{\chi_0}{1 + \left(\frac{1}{2}(c + c^+)/c_{\mathrm{sat}}\right)^2}, \quad J = \underbrace{\chi\, \mathrm{d}c\, \rho_\uparrow \left(1 - \frac{\rho_\downarrow}{\rho_{\max}}\right)_+}_{\text{advection}} - \underbrace{D_\rho(\rho^+ - \rho)}_{\text{diffusion}}, \tag{2}$$

where $\rho_\uparrow = \rho$, $\rho_\downarrow = \rho^+$ if $\mathrm{d}c > 0$, else swapped. The density update is the flux divergence $\partial_t \rho = -\sum_{\text{axes}}(J - \mathrm{roll}(J, 1))$. The extensive carriers U, V, C, M_k advect with the same flux using donor–cell intensive fractions (U/ρ, V/ρ, C/ρ, M_k/ρ).

S1.3 Growth (nutrient uptake), death

With $\sigma = (u-v)/(u+v)$, $q = (1-\rho/\rho_{\max})_+$, and dose readout $m_+ = \tanh(m_1+m_2)$:

$$g = \mathrm{d}t\, g_0\, \rho\, N\, q\,(1+\beta\sigma)\,(1+\lambda_+ m_+), \qquad g \leftarrow \mathrm{clip}(g, 0, N), \tag{3}$$

$$N \leftarrow N - g, \quad \rho \leftarrow \rho + g, \quad U \leftarrow U + gu, \quad V \leftarrow V + gv, \quad M_k \leftarrow M_k + g\, m_k. \tag{4}$$

New mass inherits the local internal state and local memory. The uptake observable is uptake $= g$. Death is homogeneous, $X \leftarrow (1 - \mathrm{d}t\, k)X$ for $X \in \{\rho, U, V, C, M_k\}$, so intensive concentrations and memory are invariant under pure turnover.

S1.4 Internal bistable identity (confined to the scaffold)

Where $\rho > 10^{-4}$ and $a > 0$, on concentrations:

$$\dot u = \frac{a}{1+(v/K)^2} - u, \quad \dot v = \frac{a}{1+(u/K)^2} - v, \qquad u \leftarrow u + \mathrm{d}t(\tau \dot u + D_{\mathrm{int}}\Delta u), \tag{5}$$

(analogously v), then clipped to ≥ 0 and re–made extensive $U = \rho u$, $V = \rho v$. This network never enters χ, c, or any cohesive term.

S1.5 Memory writing (frozen) and the two readouts

Let $\overline{\text{uptake}}$ be the entity–mean uptake. The single local experience signal is

$$\Psi = \tanh\big(k_{\exp}(N - c) + k_{\mathrm{up}}(\text{uptake} - \overline{\text{uptake}})\big). \tag{6}$$

Each intensive component, where $\rho > 10^{-4}$, obeys write / two–timescale forgetting / templating / diffusion:

$$m_k \leftarrow \mathrm{clip}\Big(m_k + \mathrm{d}t\big[\eta_w \Psi - \eta_{d,k} m_k + \eta_t(\widetilde{m_k} - m_k) + D_m \Delta m_k\big],\ -1, 1\Big), \tag{7}$$

with $\eta_{d,1}$ (fast) and $\eta_{d,2}$ (slow), and the templating reference $\widetilde{X} = \frac{1}{4}(X_{i\pm1,j} + X_{i,j\pm1})$ (periodic four–neighbour mean). Readouts:

$$\text{dose (channel 1): } m_+ = \tanh(m_1+m_2) \ \rightarrow\ \text{uptake gain } (1+\lambda_+ m_+) \text{ [in S1.3]}, \tag{8}$$

$$\text{order (channel 2): } m_- = \tanh(m_1-m_2) \ \rightarrow\ c \text{ production } (1+\lambda_- m_-). \tag{9}$$

The attractant and nutrient close the system:

$$c \leftarrow c + \mathrm{d}t\big(D_c \Delta c + s\,\rho\,(1+\lambda_- m_-) - \delta c\big), \quad N \leftarrow N + \mathrm{d}t\big(D_N \Delta N + F(N_0 - N)\big). \tag{10}$$

S1.6 Frozen parameters

Scaffold (blob `7c91b91`): $\chi_0 = 9.5$, $D_\rho = 0.07778$, $D_c = 0.68$, $s = 0.2$, $\delta = 0.06909$, $g_0 = 0.06154$, $k = 0.02588$, $F = 0.02421$, $D_N = 0.5$, $N_0 = 1.0$, $\rho_{\max} = c_{\mathrm{sat}} = 1.0$, $a = 2.0$, $K = 0.5$, $D_{\mathrm{int}} = 0.008$, $\tau = 0.20$, $\beta = 0.6$, lattice 64, $\mathrm{d}t = 0.1$. Memory (selected candidate **C1c**, frozen `FREEZE_C1c.json`): $\eta_w = 0.015$, $\eta_{d,1} = 0.35$, $\eta_{d,2} = 0.006$, $\eta_t = 0.010$, $D_m = 0.010$, $\lambda_+ = 0.25$, $\lambda_- = 0.15$, $k_{\exp} = 1.0$, $k_{\mathrm{up}} = 1.0$, $n_{\mathrm{comp}} = 2$. (C1c is the de–saturated writing correction of the IOM–00 default $\eta_w = 0.05, \eta_{d,1} = 0.03, k_{\exp} = 2.0$.)

S1.7 History coordinates

Each history applies an early and a late environmental drive of amplitudes $a_{\text{early}}, a_{\text{late}}$. The two decoded coordinates are the cumulative dose $h_1 = a_{\text{early}} + a_{\text{late}}$ (global, non–individuating) and the temporal order / asymmetry $h_2 = a_{\text{late}} - a_{\text{early}}$.

S1.8 Entity detection

Connected components (periodic 4–connectivity) of the mask $\{\rho > 0.30\,\rho_{\max}\}$; components with ≥ 12 cells are entities. Centroids use the periodic circular–mean estimator; radius of gyration and all phenotype statistics are computed on the entity cells with ρ–weighting.

S1.9 Decoding pipeline (how every R^2 is obtained)

Memory feature vector per entity: for each component m_k over the entity cells, the five order statistics $[\text{mean}, \text{std}, P_{10}, P_{50}, P_{90}]$, concatenated over $k = 1, 2$ to a **10–D** vector (entities with < 4 cells are undefined and censored). Decoder: **ridge regression**, regularisation $\lambda = 1.0$, evaluated by **grouped leave–one–history–out** cross–validation (the held–out fold is an entire history class, never a single entity, to prevent leakage). Uncertainty: donor–level **percentile bootstrap**, $n_{\text{boot}} = 3000$, resampling the grouping unit (history/seed), reported as 95% CIs. Longitudinal tracking: periodic mask–overlap matching, overlap threshold $\theta = 0.1$, evaluated **every 5 steps**, censored on track loss; the tracker never reads h_1, h_2, M, or any label. Turnover fraction M is the surviving mass fraction of the original cohort field.

2 S2. Protocol history (the nine–experiment ladder)

Each experiment below was specified by a protocol committed *before* its prospective family was generated and executed once (see the provenance table in the main text for protocol / commit / freeze / execution timestamps and hashes). HMC (material continuity): snapshot nearly sufficient, no strong history–specific hidden persistence. HSI (hidden state): generic causal attractor, not individuating (AUC 0.3–0.4). IOM–00: designed two–component memory; causal, erasable, transplantable; readout ≈1–D. MCM–00: m_{-} →attractant makes order readable (∼70/73× ablation collapse). WD–01 Phase B: grouped–evaluation correction (0.57 → 0.19); rank ≈1. Phase C: de–saturated writing (C1c); two storage coordinates at rest, causal response ≈1–D. SMC–01: h_2 frame–free dispersion; a turnover collapse later corrected. DMM–01: correction — inter–family variance, not smoothing/dilution; underpowered. H2–CERT–01: sealed kill–switch, h_2 canonical point -0.24 at $M \approx 0.19$ with a CI spanning the 0.50 threshold (retention *not established*). TCA–01: tracker–continuity audit; longitudinal re–certification of h_1.

3 S3. Branch / commit registry

experiment	result commit
HMC	25c419a
HSI	709f963
IOM–00	0ea1250
MCM–00	5841b9b

WD–01 Phase B	9a39f8c
WD–01 Phase C (seal / result)	be08738 / a07b95b
SMC–01 (seal / result)	d0f47b6 / 3dc8c35
DMM–01 (seal / result)	5bd46e7 / 49e96ad
H2–CERT–01 (seal / result)	5e1196f / 1f8f789
consolidation / observer	112aace / 6891ef8
tracker audit / post–incident	568af39 / 7afd6e4
short–paper V0 / full manuscript	a97909a / (this commit)

Main branch never modified (HEAD f3921a4); nothing pushed (origin/main e17f431).

4 S4. Sealed prospective family hashes (SHA–256, generated before selection, executed once)

experiment	seeds (dev / prospective)	SHA–256
Phase C	34001–3 / 35001–3	c6d0cd3c5b82aa122ac7e5649888fbc53ca6faa8db68e2c1b48863e87fb9e202
SMC–01	36001–3 / 36501–3	2d6986fe2499cb52fdc309f5b1867cff56cb04b4bd7884633c4c2a148d5792ef
DMM–01	37001–3 / 37501–3	923d8890ea25ab5a136078486f282ad8e07f4cdbaa125da40ef930f5bb8a00b6
H2–CERT	— / 38501–4	4265b98cacb705a7bea56a2e0a5bbe36e10fec3d977e7da6e655037189193fee

Seed families disjoint across experiments (32xxx–38xxx). "Executed once" means the prospective family was run a single time after the protocol freeze; no prospective seed was re–run to obtain a preferred number.

5 S5. Parameter grids and negative candidates

Writing–redesign candidates compared at development (Phase C): C0 (frozen baseline), C1a/C1b/C1c (dynamic–range corrections varying $\eta_w, \eta_{d,1}, \eta_{d,2}, k_{\text{exp}}$), C2a (distinct write signals). Development $\min(R^2(h_1), R^2(h_2))$ on family F_mid: C0 0.19, C1a 0.84, C1b 0.61, C1c 0.84, C2a 0.79; C1c selected for the highest stored second–dimension ratio ($\sigma_2/\sigma_1 = 0.229$) at equal top score, by minimality (a same–signal C1). Maintenance candidates (DMM Phase A, diagnostic only): templating and diffusion ablations and static memory relaxation all left the h_2 decode essentially unchanged. Negative variants were retained, not discarded.

6 S6. Gate certificates (machine–readable)

JSON gate certificates are committed per experiment: `results/wd01/wd01_gate_certificate.json`, `results/wd01_phasec/wd01_phasec_gate_certificate.json`, `results/smc01/smc01_gate_certificate.json`, `results/dmm01/dmm01_gate_certificate.json`, `results/h2cert/h2cert_gate_certificate.json`. Each records the preregistered gates and the observed values, including the gates that were *not* met.

7 S7. Tracker unit tests (TCA–01)

(i) Selection audit: the historical convenience rule is arg max size, reselected per frame; five hard switches occurred on seed 38501. (ii) Periodic translation invariance: whole–world shifts (including boundary–straddling) leave size and memory statistics bit–invariant. (iii) Sensitivity: h_1 decodes 0.91/0.89 from largest / longitudinal trackers (tracker–independent; canonical pipeline). (iv) Hold–out: untouched seeds 38502–38504, 36/36 track survival, h_1 deep $R^2 = 0.89$ CI $[0.84, 0.96]$, h_2 point -0.24 CI $[-0.78, 0.32]$ (canonical committed pipeline). Specification: `docs/audit/TCA_01_TRACKER_SPEC.md`.

8 S8. Seed–family results (h_1 turnover) and the re–audited intervals

DMM pool and H2–CERT sealed families (historical, non–canonical pipeline; retained for provenance only): DMM pool h_1 high across $M = 1.00 \to 0.36$; H2–CERT sealed (4 seeds): $0.94 \to 0.93 \to 0.93$ over $M = 1.00 \to 0.44 \to 0.21$. Longitudinal hold–out (untouched seeds): initial $R^2 = 0.78$ CI $[0.56, 0.94]$, deep 0.89 CI $[0.84, 0.96]$ — the paper's primary quantitative result (canonical committed pipeline). Re–audited comparators (grouped bootstrap, $n_{\text{boot}} = 3000$): transplant h_1 0.61 CI $[-0.69, 0.87]$; in–place h_1 0.50 CI $[-1.39, 0.92]$; memory–inert / erased deterministic -0.19; memory vs. size+mass 0.93 vs. 0.64 (CIs overlap); longitudinal h_2 deep point -0.24 CI $[-0.78, 0.32]$ (below 0.50; *not established*). Full methodology: `STATISTICAL_REAUDIT_V4.md`; canonical pipeline `reproduction/`.

9 S9. Reproduction commands and data availability

Environment: python3, numpy, scipy, matplotlib; `PYTHONPATH=$REPO:$REPO/results/wd01_phasec`. Per experiment, resumable runners are committed beside their data: `results/wd01/run_diag.py`; `results/wd01_phasec/{dev_runner,prosp_runner,causal_runner,causal_inplace}.py`; `results/smc01/{fie` `results/dmm01/{phaseA,pooled,prosp}.py`; `results/h2cert/{pilot,cert}.py`; the tracker hold–out reruns deterministically from the H2–CERT manifest seeds 38502–38504. Viewer: `scripts/observe_droplet.` `--preset story`. *Data availability.* All raw arrays, JSON certificates, sealed manifests, and figure generators referenced here are committed to the branch `paper/organizational-memory-boundaries-01` of the repository `emergent-dynamics-lab`; every reported number derives from a file in that commit and no analysed run was excluded from the reported statistics.

10 S10. Raw–data and figure registry

Raw: `results/{sc_mcm,wd01,wd01_phasec,smc01,dmm01,h2cert,observer}/*.pkl,*.json`. Figures (paper): Fig. 1 fig1_model_ladder; Fig. 2 fig2_h1_causal_v3 (predicted–vs–observed with bootstrap CI); Fig. 3 tracked_3panel; Fig. 3b fig3b_longitudinal_certification (primary result); Fig. 4 fig4_order_m_minus; Fig. 5 phasec_figure; Fig. 6 h2cert_figure; Fig. 7 paper_claim_ladder (corrected historical summary). Each figure traces to a committed generator script.